

Longitudinal Confidence Modelling for Motor Subtype Classification in Early Parkinson's Disease

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What we were doing before

We were trying to sort PD patients into "slow / moderate / fast progressor" groups by running K-Means clustering on their decline rates (motor, cognitive, autonomic). The algorithm decided where the group boundaries were — nobody outside our project has ever validated that these three groups are clinically real.

The problem: our research guide flagged that any subtype we invent ourselves needs a clinician to confirm it's actually a meaningful medical category before we can treat it as ground truth. Getting that kind of clinical sign-off — especially from a specialist abroad — isn't realistic on our timeline.

Our own data backed this concern up: when we tested a different clustering method (HDBSCAN) that doesn't force a fixed number of groups, it couldn't find three well-separated clusters at all — it called 39% of patients "unclassifiable." That's the data itself telling us slow/moderate/fast is a fuzzy, invented boundary, not three real groups.

What we're changing

We stop inventing the category. We use one that already exists.

Instead of predicting a group we made up, we predict **TD (tremor-dominant) vs. PIGD (postural instability / gait difficulty) vs. indeterminate** — the standard three-way motor subtype used across Parkinson's research worldwide.

This isn't something we cluster or guess. It's a **published formula**: take a patient's tremor-related exam scores, take their gait/balance-related exam scores, compute a ratio. The ratio tells you the category. This was developed by Jankovic (1990) and formally validated for the modern rating scale by Stebbins et al. (2013, *Movement Disorders*) — it's the accepted research-standard method, used in PPMI and most PD subtype studies.

Why this solves the verification problem: we're not asking a clinician to bless a new idea. We're applying a peer-reviewed formula correctly to data we already have. That's a much smaller ask than validating an invented category — and it may not require live clinician review at all, just correct citation and correct calculation.

What stays exactly the same

Same dataset (PPMI, same 229 patients, same multi-visit structure)

Same core research question: can we reach a confident classification **earlier**, instead of waiting a year+ of visits like a doctor currently does?

Same tools: conformal prediction, our confidence-curve idea, prediction stability tracking

Same eventual goal: a system that tells a doctor not just "this patient is likely PIGD" but *how confident* that call is, and how that confidence builds up visit by visit

What's new: the "transition risk" angle

Published research shows TD/PIGD labels aren't fixed — a meaningful share of patients (roughly a third to half, in published PPMI-based studies) get reclassified within a year or two as their disease evolves. Existing papers just note this as a limitation. **Nobody is building a system that tells you, per patient, how likely their current label is to hold vs. flip.**

That's our actual contribution: not just "this patient is PIGD," but "this patient is PIGD, and our confidence in that is high/low, and here's how many more visits it would take to be sure." This uses tools we already have (conformal prediction, stability index) — we're just pointing them at a target that's clinically standard instead of one we invented.

What needs to happen next

Confirm we have the right data. The TD/PIGD formula needs *individual item scores* from UPDRS Part II and III (specific tremor items, specific gait/balance items) — not just the part totals. Need to check whether the joined dataset kept item-level columns or only totals.

Compute TD/PIGD/indeterminate per patient, per visit, using the Stebbins formula.

Re-run our existing pipeline (classification + conformal confidence + stability tracking) on this label instead of the old K-Means one.

Watch for class imbalance — TD patients heavily outnumber PIGD ones in PPMI-like cohorts, so we'll need to account for that (weighting, or reporting performance per class).

Bottom line: same infrastructure, same core idea, one swapped target variable — and the swap removes the clinical-verification bottleneck instead of just relocating it.